

tions but contact your phone's manufacturer website if the instructions do not fit your particular model.

- Complain to the service to offer more than just SMS for 2FA! The more complaints they get, the more likely they might be to offer more secure options.

Additional 2FA Concerns

Remember the purpose of 2FA - it is a secondary authentication method, invoked after the primary authentication method has succeeded. You should only acknowledge a 2FA request if you personally have just successfully completed the primary authentication. An unsolicited 2FA request means someone has your password, and is in the process of trying to log in as you. This is why 2FA exists, to help protect your user account from attacks involving your password.

If you experience any such irregularities, please bring it to the attention of the Security Team. For more information, review the handbook regarding Security Awareness

Slack

There is a lot of information pertaining to Slack, as it is a critical part of GitLab's communication. See the Slack handbook page.

Zoom

There is a lot of information pertaining to Zoom, as it is a critical part of GitLab's communication. See the Zoom handbook page.

The Google Calendar invite is the single source of truth for the Zoom link. Avoid linking the Zoom link in Google Document agendas since those may quickly become out of date. If the zoom link changed around the start of the meeting it is OK to have it in there temporarily.

Google

Need a new document or new slide deck quickly? Use shortcuts like `<https://doc.new>` or `<https://slide.new>`. The full list (not even restricted to Google products) is available at `<https://whats.new/shortcuts/>`.

Google Docs

Google Docs Pro Tips

1. Quickly create a new Google Doc in chrome: type "docs.new" in the chrome address bar. Likewise "sheets.new" for a Google Sheet etc...
1. While in a document with many other editors, click on the image or icon of any user at the top of the document to move focus to their cursor and what they are typing in the document. This is great when someone is speaking about something they are typing on a video call and you are not sure where they are in the document.
1. Quickly find all action items assigned to you with a search for `followup:actionitems`.
1. To add additional rows to a numbered list in a Google Document, press Return to add additional spaces under the existing numbered list in the document, highlight those spaces, and click the numbered list icon in the formatting bar (or press Command · + Shift + 7 on a Mac) to

create a numbered list from the spaces. See video on how to add more rows to a numbered list for a quick tutorial.

1. Type the @ character to quickly embed today's date, other document titles, references to people, and more; without clicking anything (a menu will pop up). See details in the Smart Canvas blog post and the Add items with the @ menu support doc.

- Emoji live-search can be accessed by typing @:, or shorter using the : character. Start typing to search for an emoji and press enter to insert.

1. Pageless docs are great for documents you don't intend to print, and is the preferred format at GitLab. You can switch in File > Page setup.

See also Live Doc Meetings for more Google Doc tips.

Google Analytics

Google Analytics (GA) is an essential tool for making data-driven decisions. It receives data from both about.gitlab.com and docs.gitlab.com websites. Read through the Inbound Marketing Handbook for more information on GA.

For example, you can look at the GA data to analyze how visited is a certain page, in a period of your choice.

You can also look at the GA referrals data to understand where the users are coming from and where they go when they leave a certain page.

To see the data for a specific page:

- Open GA, and expand Behavior on the sidebar
- Click Site content > All pages
- On the top-right, adjust the period of time you'd like to analyze
- On the middle of the page, look for a search bar and paste the URL you'd like to analyze (without https://) and click on the magnifier button to search:

Note that you can use the search tool with:

- A full URL, which will return results for that specific URL
- Part of an URL, e.g., /2017/, which will return the results for all the blog posts published on 2017
- The higher directory on the file tree, which will return the results for a range of URLs in that tree.

E.g., docs.gitlab.com/ee/ci/ will return the results for all the range of pages contained in the /ci/ directory

- GA will output the data about the page (or range of pages) you searched for, including pageviews, unique pageviews, and other data:

To find the referrals for a certain page, continue from the steps above.

- Click on one of the website links to look at the data for a page of your choice
- Just above the graph, click Navigation summary
- GA will output the referrals, including Previous Page Path and Next Page Path:

Set your picture in Google

Optional: Set your picture in Google so that your picture will show where you are in a Google document (vs showing just your first initial). This will allow others to more easily follow a discussion when meeting attendees move around in a document.

Add name pronunciation in Google

Consider adding a phonetic pronunciation of your name and/or a pronunciation recording in your Google profile.

- Click on your profile icon on the top-right of any Google account page (i.e. Google Docs, Google Sheets)
- Select "Manage your Google account"
- On the left-hand side panel, select "Personal info" >> on the center of the page, go to "Name & pronunciation"
- Add a recording or a phonetic pronunciation e.g. Rochana (Rosh-ah-na)

Google Calendar

Finding a time

Please make use of the Find a Time tab in Google Calendar, especially when scheduling events with teammates in other parts of the world:

Find a Time presents a new or existing event's time for all participants, adjusting for time zones as appropriate.

To use Find a Time:

1. Create a new event or modify an existing event.
1. Click the "Find a Time" tab. Invited guests will be presented in the availability table and represented by a column.
 - Areas outside of someone's working hours (9:00 AM - 5:00 PM by default) are represented in light grey.
 - Guests who are optional will not appear in the availability table by default. You can add them by checking their name in the "Guests" area on the right hand side.

For meetings spanning across multiple time zones and with external parties, Time & Date Calculator can help determine the best time to schedule.

GitLab availability calendar

The GitLab Availability Calendar has been deprecated to allow for GitLab to scale effectively. We have created tools and tips for managing your time off.

Setting working hours & location

To help other team members to schedule meetings with you, you can set your normal working hours & location in Google Calendar. The preferred times will show up when someone tries to schedule a meeting with you.

To set your working hours, go to Settings -> General -> Working Hours & Location. Working location can be set here or on your calendar directly by clicking on the pill in the All day row. In Settings there's a helpful Copy to all option as well.

Since all GitLab Team Members work remotely Home can be an ambiguous choice as folks may travel and it doesn't provide any context on timezones. If you'd like, you can choose Somewhere else (in Settings) or Edit pencil -> + -> Another location (from the pill) to set this to something such as City, State, Country (GMT-) and keep it up to date when traveling or timezones shift, which can help folks with planning and understanding timezones.

GitLab Team Meetings calendar

The GitLab Team Meetings Calendar is available to all team members and can be found in your calendars list after it's added. To add the GitLab Team Meetings Calendar to your list of calendars:

1. Go to your calendar
1. Scroll the left sidebar down to Other calendars
1. Press on +
1. Select Subscribe to calendar
1. Enter in the search field `gitlab.com6ekbk8ffqnkus3qpj9o26rqejg@group.calendar.google.com` and then press enter on your keyboard

Please reach out to a People Connect Team member if you have any questions. NOTE: Please do NOT remove any meetings from this calendar or any other shared calendars, as it removes the event from everyone's calendar.

You can find the details for the Company Calls, Group Conversations, 101s, and other teams' meetings here, so you can attend a different teams' meeting and ask questions, learn about what they're working on, and get to know the rest of the GitLab Departments and teams.

These meetings are open to everyone in GitLab.

If you are creating a new team meeting, please add it to the GitLab Team Meetings calendar

Please reach out to the People Connect Team in the people-connect Slack channel with any questions, requests or changes to the GitLab Team Meetings calendar.

Adding an event to the GitLab Team Meetings calendar

1. Create your calendar invite
1. Add your Zoom link
1. Add an agenda or relevant content
1. Under Add Guests, add GitLab Team Meetings in addition to anyone else you want invited (if you want the entire company invited please use `everyone@`).

{{% note %}}

This will appear like adding a room, which is expected

{{% /note %}}

Managing invite responses

If you are familiar with queries in Gmail, add a filter to remove invites responses from your inbox with the following query:

.ics subject:("invitation" OR "accepted" OR "rejected" OR "updated" OR "canceled event" OR "declined") when where calendar who organizer

You can also create a filter to remove ALL invite responses from your inbox with the following search terms:

- Subject: "invitation" OR "accepted" OR "rejected" OR "updated" OR "canceled event" OR "declined"
- Has the words: .ics

Then click Create filter:

- Skip the Inbox
- Apply the label (of your choosing)

Modifying events

Please click 'Guests can modify event' so people can update the time in the calendar instead of having to reach out via other channels.

You can configure this to be checked by default under Event Settings.

Notifications

You can change the default notification settings by calendar (add or remove notifications for all-day events, add a second default notification to all events...). This is accessible through Settings -> Settings for my calendars -> pick the calendar -> Event notifications and All-day event notifications.

If you change the default notifications, all existing events will inherit that as well (unless their notifications have been customized).

Restore deleted calendar items

(This assumes you are using Google's new Calendar).

When you have accidentally deleted something from the Team Meetings calendar, you can recover it by:

- Go to Google Calendar and click the gear icon at the top left of your screen.
- Choose the Trash.
- Make sure you are on the correct calendar, by clicking on the name of the calendar in the left sidebar.
- Hover over the item you'd like to restore and click the arrow to "Restore".

Show declined events

We recommend enabling the 'Show declined events' setting if you are unable to attend a meeting but will still contribute to the agenda or attendance list asynchronously. Use this setting if it's helpful for you to see declined meetings in your calendar view for any other reason.

- Navigate to Event settings
- Check the box for 'Show declined events' under 'View options'

Sharing

We recommend you set your Google Calendar access permissions to 'Make available for GitLab - See all event details'. Team member calendars should not have access permissions set to 'Make available to public' due to the risk of sensitive data exposure and zoombombing.

Consider marking the following appointments as 'Private':

- Personal appointments
- Confidential & sensitive meetings with third-parties outside of GitLab
- 1-1 performance or evaluation meetings
- Meetings on organizational changes

There are several benefits and reasons to sharing your calendar with everyone at GitLab:

1. Transparency is one of our values and sharing what you work on is in line with our message of "be open about as many things as possible".

1. Due to our timezone differences, there are small windows of time when our availabilities overlap.

If other members need to schedule a new meeting, seeing the details of recurring meetings (such as 1-1s) will allow for more flexibility in scheduling without needing to wait for a confirmation from the team member.

This speaks to our value to be more efficient.

When setting up your Google Calendar be sure to set your working hours.

If you'd like to share your calendar with e.g. your partner you can use the 'Share with specific people' feature and set the permissions to 'See only free/busy (hide details)':

Speedy meetings

Enable speedy meetings to automatically provide a buffer at the end of events you schedule. This thoughtfully allows participants with back-to-back events the opportunity to use the restroom or grab a cup of coffee without being late to their next function.

World clock

Add as many time zone world clocks as you wish by, in Google Calendar, going to Settings -> World Clock in order to see team members' local times.

You can also use sites like TimeAndDate to convert times to/from UTC for example.

Time zone

Check Display secondary time zone and select (GMT+00:00) Coordinated Universal Time (UTC). This enables team members to standardize on a single time zone in communicating when meetings take place.

Google Cloud Platform

See the Sandbox Cloud page for a listing of cloud resources and how to gain access to them.

Google Drive

First, an important message - Don't use Google Drive/Apps (unless you have to)

We would be remiss if we didn't start this section off with this IMPORTANT message: Your default storage place for information that needs to persist and be available to others in the company should be ON THE WEBSITE/IN THE COMPANY HANDBOOK and not in Google Drive and Google Apps files!! This is from the top.

This is how we operate, because Google Docs/Apps can only be found and contributed to by team members, and not by users, customers, advocates, future employees, Google handbook searches, or developers.

Do not link directly to Google Drive/Apps

Having said that, there is content which doesn't make sense to be created on the website directly (e.g. large collections of data in tables, spreadsheets for calculations, etc) or for which Google Drive storage makes sense.

When directing folks to these files in Google Drive please include name of the file in the handbook so that team members can search for it in Google Drive. If you link directly to the URL, people from outside the organization can request access, creating workload and the

potential for mistakes.

Keeping it organized

It is important that we not just throw files into random or general places in the shared Google Drives.

Doing so makes it harder for others to find and work with the content.

Here are some guidelines to organizing the Google Drive content:

- First by department (e.g. strategic marketing)
- then by subject (e.g. analysts relations)
- then by sub-subjects as deep as necessary (e.g. Gartner -> 2018 ARO MQ)

Using Google Drive

For starters, when your GitLab Google company account is created you automatically get a Google Drive with unlimited storage allocation in your own "home" directory (called My Drive).

You can get to it by:

1. (optional) Login to your GitLab account in your browser (if you are using Chrome)
1. Open your web browser to <<https://drive.google.com>>
1. If you're not already logged in as your GitLab account (Chrome users should be) then login to Google using your GitLab account
1. This will take you to your Google Drive (called My Drive) which is like your home directory. If you create Google files using Google Apps and don't specify where to store them, they will be put in this home directory.

This is great for storing your own working files.

As already stated, this should never be the final resting place for shared files that are meant to be used by the rest of the company (or beyond).

Existing GitLab Google Drive repositories

There are a few Google Drive repositories of GitLab shared files (there might be more, please add if not listed here):

- UX Research Drive - This houses all findings from Customer Discovery Meetings and raw, confidential materials from customers shared with Design, UX Research, Product and Customer Success. Distilled findings from User Research are stored in the UXRI Insights repository.
- GitLab Marketing Drive - This houses all shared files from the entire Marketing organization. The best practice is for sub-organizations to have their own directory inside this space (e.g. Strategic Marketing).
- Sales Drive - This houses all the shared files from the Sales organization. The best practice is for some sub-organizations to have their own directory inside this space (e.g. Customer Success).
- GitLab Alliance Drive - This houses all shared files from the entire Alliance organization. The best practice is for sub-organizations to have their own directory inside this space (e.g. Partner Discussions).

How do you use these? You don't have to remember these URL's.

To add these links to your Google Drive My Drive directory, do the following:

1. Make sure you are logged into your GitLab account in Google Drive in your browser
1. Open the link of interest (from above) to go to that directory
1. Find the directory path across the top (under the "Search Drive" field)
1. Find the name of directory in that path that you want to add to your drive (e.g. Sales)
1. Click on the down arrow next to it
1. From the resulting pop-up menu, select "Add to My Drive"
1. From now on you can get to that directory by first going to your drive (<<https://drive.google.com>>) and then opening that link

Adding Google Drive to your Mac

To really make your Google Drive easier to access, you can have your Google Drive show up on your Mac Finder as a regular drive.

With this it is easier to store and view files such as videos, analyst reports (PDFs), etc.

Here's how to do this:

1. Make sure you are logged into your GitLab account in Google Drive in your browser
 1. Go to your Google Drive (<<https://drive.google.com>>)
 1. Click on the "Settings" icon (Gear) to the right of the search field
 1. From the resulting menu, select "Get Drive for desktop"
 1. It might pull up a new page/tab and use your personal login.
- If it does this you won't see "Download & install Drive for desktop" on the page.
Switch to your GitLab account.
1. Download and install

Google Forms

Use these GitLab branded form templates when creating internal or external surveys or forms. Make a copy of the form and only edit the copy; do not edit the template itself.

Review what data privacy means at GitLab. Help participants make informed decisions by applying the following guidelines:

- For external forms: State that any Personal Data collected will be processed in accordance with the GitLab Privacy Statement.
- For internal forms: Ensure that the purpose for collecting data is stated under our Team Member Privacy Policy and that this purpose is made clear to Team Members.
- Don't make Personal Data form fields mandatory unless they're absolutely necessary to achieve the processing; any inessential Personal Data collected should be through a voluntary form field.
- Only process Personal Data for the reason in which it was initially collected e.g., if the Google Form collected traveler information, don't then use that Personal Data to create a marketing list.

Google Mail (Gmail)

Auto-advance

If you use the archive function, you normally return to your overview.

With Auto-advance you can select whether to advance to the next or previous message.

"Auto-advance" can be enabled from the Advanced section under Settings.

This reveals the Auto-advance settings in the General section under Settings.

The default setting of showing the previous (older) message is usually preferred.

Email signature

-Set up an email signature which includes your full name and job title so people can quickly know who you are and what you do.

-It is also an option to add your personal pronouns to your email signature.

Example

Note: You can copy and paste the template below to use it in your own signature.

```
<span style="font-family: serif;font-size: small;display: block;">Alex Doe (they/them)</span>  
<span style="color: 999999;font-family: sans-serif;font-size: small;display: block;">Frontend  
Engineer | GitLab</span>
```

```

```

Email signature if employed by our German entity

During onboarding you may have been asked to set up your GitLab Gmail and your email signature using the following example as a guideline. If you are employed by our German entity, you will need to add some company related details to the end of your email signature as set out here (accessible internally to GitLab only).

Add company details to certain company communications if employed by our German entity

Certain company details are required to be added as a signature to the forms of communication listed below, where that communication is going from a team member employed by our German entity GitLab GmbH to outside of the company, and relates to GitLab's business activities:

- order forms
- physical mail
- fax
- postcard
- email
- SMS
- Twitter
- Zendesk
- other electronic communications

If you are sending such a communication, externally, please look at the setting of the relevant application and add the company details below to the end of the communication. If you are in doubt, please do add the details. If you are unable to add this information for any reason, please alert legal-employment@gitlab.com.

If you are employed by our German entity you may have been asked during onboarding to set up your signature wherever it's required to be used and to include certain company details, and if you have not been, please do so now, as set out here (accessible internally to GitLab only).

Filters

Apply label on mention

It might be useful to add a Gmail filter that adds a label to any GitLab notification email in which you are specifically mentioned, as opposed to a notification that you received simply because you were subscribed to the issue or merge request.

1. Search for from:(gitlab@mg.gitlab.com) "you+have+been+mentioned+on".
1. Click the down arrow on the right side of the search field.
1. Click Create filter with this search.
1. Check Apply the label: and select a label to add, or create a new one, such as "Mentioned".
1. Check Also apply filter to matching conversations.
1. Click Create filter.

Apply label when assigned as reviewer

You can create a Gmail filter that adds a label to any GitLab notification email in which you are assigned as a reviewer:

1. Search for from:(gitlab@mg.gitlab.com) "(Reviewer) (Firstname Lastname)".
1. Click the down arrow on the right side of the search field.
1. Click Create filter with this search.
1. Check Apply the label: and select a label to add, or create a new one, such as "Assigned".
1. Check Also apply filter to matching conversations.
1. Click Create filter.

Apply label when MR is merged

You can create a Gmail filter that adds a label to any GitLab notification email in which an MR has been merged:

1. Search for from:(gitlab@mg.gitlab.com) "Merge Request"+"was merged".
1. Click the down arrow on the right side of the search field.
1. Click Create filter with this search.
1. Check Apply the label: and select a label to add, or create a new one, such as "Merged".
1. Check Also apply filter to matching conversations.
1. Click Create filter.

Apply label to all GitLab-generated emails

GitLab issues and merge requests can generate a lot of email notifications depending on your settings and how in-demand your attention is.
It can be useful to apply a label to these generated emails and move them out of your immediate inbox.

1. Search for from:gitlab@mg.gitlab.com.
1. Click the down arrow on the right side of the search field.
1. Click Create filter with this search.
1. Check Skip the Inbox (Archive it).
1. Check Apply the label: and select a label to add, or create a new one, such as "GitLab.com".
1. Check Also apply filter to matching conversations.
1. Click Create filter.

You can learn more about how to use Gmail filters to organize your inbox in Productivity Hack video.

To import downloaded filter export go to Gmail => Settings => Filters and Blocked Addresses => Import filters.

Keyboard shortcuts

Keyboard shortcuts only work if you've turned them on in Gmail Settings.

Steps below:

- In "Settings" scroll down to the "Keyboard shortcuts" section
- Turn Keyboard shortcuts "on"
- Scroll down and Save Changes

Here are some shortcuts you can use

Split screen

List your inbox and preview mails in one view with this configuration change:

- Cog/settings top right of inbox.
- Settings option.
- Inbox tab
- Reading pane: enable
- Select a position for the reading pane, Right of Inbox or Below Inbox
- Save changes
- Reload inbox

Inbox Zero

To utilize Gmail to its full potential, consider adopting the Inbox Zero strategy. It's the same way Google employees use Gmail.

There's also an internal training recording about this. It goes into more details and more "power user" focused, covering keyboard shortcuts, etc.

- Video (It's set to private, so log in with GitLab Unfiltered to watch it: upper right corner > click on your profile picture > switch account > GitLab Unfiltered)
- Meeting notes

Disable Google Meet and Chat in Google Mail

The integration for Google Meet and Google Chat is enabled by default. It can consume too much space in the left menu listing mailbox folders. In order to disable the integration in Google Mail, navigate to Settings at the right top, See all settings, Mail and Chat and select the following:

- Chat: Off
- Meet: Hide the Meet section in the main menu

Save the changes and wait for Google Mail to reload.

Google Slides

Use this general GitLab Slide Template when creating slide decks for internal and external use. Make a copy of the slide deck and only edit the copy; please do not edit the template itself. To avoid potentially editing the source deck, you can go to the Template gallery and click on the GitLab Slide Template vYear-Month to quickly make a copy and begin a new deck. Ensure that all pages in the slide deck are numbered, with the title page always being number 1.

Sales slide decks

Slide decks are available in Highspot in GitLab's Official Sales Deck Library.

Google Jamboard

Jamboard is a collaborative whiteboarding platform, accessible at <https://jamboard.google.com/>. The results are persisted in Drive and are shareable like any other object.

Google Chrome

- Search and find things by typing words from document/issue/etc page titles into the urlbar (anything that appears in the page title). The search is "best effort" but surprisingly efficient at finding that doc or issue you've accessed recently.
- "Search engines" help even more! Typing "drive" and pressing Tab, for example, searches in Google Drive. To discover and customize them, go to Settings > Search engine.
- Practice Chrome keyboard shortcuts. Here are a few common helpful shortcuts:
 - Jump to the address bar: cmd l
 - Create a new tab cmd t, close a tab cmd w, undo close tab cmd shift t
 - Open new incognito window: cmd shift n (e.g. to reproduce a UI bug, or login problem)
 - Search all opened tabs: cmd shift a
 - Hide the bookmarks bar: cmd shift b (e.g. for screenshots, or shared screen presentations)
 - Open the developer tools: cmd option i (e.g. for debugging website errors)
- You can organize tabs into tab groups, assign names and custom colors, and collapse or reorder them. Moving tabs between groups also works across browser windows.

- Logging in to the browser (with your GitLab Workspace account) preserves your profile across installations. This is helpful to migrate bookmarks, extensions, and configuration when refreshing your laptop.

Chrome Performance Settings

1. Enable the memory saver. This will pause inactive tabs and reduce memory consumption.
1. Enable the energy saver. This can help extend battery life by telling Chrome to reduce background task activities.

SECTION: tools-and-tips/git.md

You know what git is. ·

Making sense of refs

- Get the current branch name
- `git rev-parse --abbrev-ref HEAD`

Listing files

- List files from the HEAD commit
 - `git diff-tree --no-commit-id --name-only -r HEAD`
- List files in the current working tree
 - `git diff --name-only`
- List files changed from master
 - `git diff --name-only master`

Finding commits

- Find the merge commit for the given commit
 - `commit=$0 git rev-list $commit..HEAD --ancestry-path | grep -E $(git rev-list $commit..HEAD --first-parent | paste -s -d "\|" -) | tail -1`

Pushing commit

- Push your commits, but skip executing CI pipeline
 - `git push -o ci.skip`

SECTION: tools-and-tips/linux.md

Linux is allowed as an alternative to an Apple Mac running macOS. Historically this was only allowed for the engineering team, but anyone in GitLab can use Linux. Just bear in mind it is assumed you are capable of self-support when running Linux - there is a linux channel in Slack where one can exchange tips and tricks, but it is not intended as an official helpdesk resource.

The following is a guide to go over the basics of what is recommended for installation to get you up and running as quickly as possible.

- Not everything listed here may apply to your use case.
- This document will not cover the customization desired for your own personal configuration and setup of your desktop/window manager/etc.
- This guide is intended to be very generic. Be aware package names listed here may be named differently on your distro of choice.
- And as always, there are multiple ways to do many things, so if you have a better preference or are more familiar with another method, feel free to do so and contribute back to this document.

Basic Setup

Outside of the basics listed here for all laptop and desktop systems, there are a few additional steps required for Linux. Currently, the recommended laptop for Linux is Dell running Ubuntu, which is discussed in detail here. These instructions assume that setup, so for alternate choices in hardware or Linux distribution it is recommended you use this as a general guideline but adapt as needed.

Dell maintains its own Ubuntu distribution. While not needed to use Linux on Dell, it does seem to offer direct support for various chipsets and hardware components. This means that for a stock Ubuntu (or other distribution) install, generic choices are often made by the installer, and if you wish to take advantage of drivers that better support the hardware, you may have to install and configure drivers for certain components. The Dell Ubuntu distribution removes that burden.

To summarize our install process to meet GitLab standards, we will go the following steps (remember these steps were current as of December 2019):

- Boot up the Dell and begin the installation of Ubuntu on the screen.
- When you reach the screen that gives you the option to create recovery media, insert a USB thumbdrive and do so.
- Continue with the installation until complete.
- Immediately back up some key files created during the installation process. If you decide to switch to generic Ubuntu (for example to upgrade to a newer Ubuntu version), you will need these files.
- Reboot and using the recovery media, restore the system but select the option to encrypt the hard drive.
- Complete the installation as before.

Initial Installation

- Note: Before installation, make sure you have a few USB thumbdrives on hand. It is highly recommended you use modern thumbdrives that are compatible with UEFI booting (most modern ones are).
- Boot up the Dell Laptop. You will reach an Ubuntu System Configuration screen. Under "Welcome" select your language and press Continue.
- You will be presented with a "Terms of Service" to review. Click the checkbox at the bottom and click Continue.

- On the "Keyboard layout" screen, make your selection and click on Continue.
- On the "Wireless" screen, choose the network, enter its password and click on Continue.
- On the Where are you? screen you enter your location to determine your time zone. Click Continue.
- On the Who are you? screen you enter in your computer name, username, your password. Click Continue.
- You will be given an opportunity to make a backup copy of recovery media. It is highly recommended you do so, even if you intend to wipe and reload the entire operating system from scratch. In the event of complete failure, you can at least recover the system. For our purposes of encrypting the hard drive, it is a must. Do this on a USB drive and make sure you label it "Recovery". Follow the prompts and press Continue when complete.
- The system will reboot.
- When it boots up, log in. If asked if you wish to "Help improve Ubuntu", select the option of "No, don't send system info" in case crash information is sent that contains proprietary work information. If it tries to update the system with patches, fine - but you do not have to update yet if you don't want to. We are getting ready to rewipe anyway.
- Insert a blank USB drive, as we need to back up some data before we encrypt the main hard drive, just in case the encryption process fails.
- The USB drive will auto-mount, usually to a location like `/media/<username>` where `<username>` is the name you log in with.
- Hold down the Ctrl key and press t to open a terminal window. Type in the following:

```
$ df
```

- This is mainly to confirm the USB drive location, in my instance this was `/media/mloveless`.
- Create a couple of directories to store backups of some critical information we might need:

```
$ sudo mkdir /media/mloveless/bkup
```

```
$ sudo mkdir /media/mloveless/bkup/apt
```

- We need to make a backup copy of all information regarding sources for updates to our system to our USB drive:

```
$ sudo cp -R /etc/apt/sources.list /media/mloveless/bkup/apt/
```

- Next we need to backup and copy signing keys used for verifying packages for future installation and update processes:

```
$ sudo apt-key exportall > apt.keys
```

```
$ sudo cp apt.keys /media/mloveless/bkup
```

- Unmount and remove the USB drive, and label it "Backup". We're now ready to encrypt the Dell hard drive.
- Insert the USB drive labeled "Recovery" and boot off of that USB drive. You may have to enter the BIOS and specifically choose to boot off of it. Instructions vary by Dell laptop, it usually involves pressing F2 or F12 during the boot process. Search on the Dell website for more information, but essentially you want to select an option that allows you to boot off of the USB drive.

- Once you boot up, you will be given some options for recovery. The language has changed for this, but the option you are looking for is a complete wipe and reload of the system. As you move forward, installation will continue, similar to the regular installation above. The main difference is you are looking for an option to encrypt the hard drive. Do so, and reinstall as normal. You will be repeating the steps from above.
- On restart, you will be prompted for your password for the encrypted drive, and you will log in as normal.
- If all goes correctly, you will end up at a normal Ubuntu desktop and can jump down to the Final Installation Steps below.
- If there is a problem where the recovery fails and you are unable to encrypt the drive (or you wish to upgrade the system to a more recent version of Ubuntu), you will need to proceed with the Alternate Installation steps below. Otherwise, feel free to move on to the General Applications section.

Alternate Installation

In some cases the "recovery" process when trying to encrypt the hard drive has failed for other team members, in other cases they decided they wanted to use a more recent version of the Ubuntu operating system than what the Dell Ubuntu version is. Either way, you can follow these steps. These steps assume you have the "Backup" USB drive created during the Initial Installation instructions.

If you must configure a fresh Ubuntu install on the Dell yourself, you may not be able to perform any steps involving the "Backup" USB drive. In this case, ignore these steps, but all other instructions remain applicable.

You will need a copy of vanilla Ubuntu copied onto a UEFI bootable USB drive (labeled "Ubuntu", and this will be a fairly straight-forward Ubuntu install. The instructions assume Ubuntu 18.04 or 20.04 LTS, although you can do the same with a later version, including non-LTS versions.

- Insert the USB drive labeled "Ubuntu" into the USB port and boot. Press F12 when you see the Dell logo. You will reach a boot screen which lists the following (if you do not reach the boot screen, try restarting again and hitting F12 quickly and repeatedly after the logo appears). Select your UEFI-compatible USB drive. This is important: If you do not use a UEFI USB device for your initial load, it could trick your install process into a non-UEFI style installation, rendering your laptop unbootable.
- Once booted up, select Install Ubuntu.
- On the "Keyboard layout" screen, make your selection and click on Continue.
- On the "Wireless" screen, choose the network, enter its password and click on Continue.
- On the "Updates and other software" screen, make sure the "Normal installation" is checked. There is no need to select the "Install third party software for graphics and Wi-Fi hardware and additional media formats" unless you decide to take advantage of additional drivers for your hardware (which you can do later if you wish). Click on Continue.
- On the "Installation type" screen, click on "Erase disk and install Ubuntu", then click on "Encrypt the new Ubuntu installation for security". "Use LVM with the new Ubuntu installation" will be checked automatically, which you can leave it. Click on the Install Now button.
- On the "Choose a security key:" screen, enter a strong password and confirm it. **DO NOT FORGET THIS PASSWORD.** If you do you will have to reinstall as you will be unable to boot. Click on the Install Now button. After a brief wait a "Write the changes to disks?" window will appear to confirm you are installing the partitions, click on Continue.

- On the "Where are you?" screen select your timezone and click Continue.
- On the Who are you? screen, you will be prompted for your name, the name of your computer, a username, a password, and your password a second time to ensure it is correct. The button next to "Require my password to login" will be selected, make sure it remains selected. For convenience sake, you may wish to make your username match your GitLab username, but this is not required.
- The install process will take a while, anywhere from a few minutes to maybe an hour, depending on your connection speed. You will eventually reach an "Installation Complete" window. Click on Restart Now. After the restart, remove the USB drive.
- On restart, you will be prompted for your password for the encrypted drive, and you will log in as normal.
- After you have reached the Ubuntu Desktop, you will have a few more steps to perform.
- Edit /etc/apt/sources.list, remove the `#` symbol to uncomment the line that reads:

```
deb https://archive.canonical.com/ubuntu bionic partner
```

- Insert the USB drive labeled "Backup" and perform the following action:
- Enter the following commands from a terminal window (the instructions below assume your username is mloveless, change to your username:

```
$ sudo apt-key add /media/mloveless/bkup/apt.keys
```

```
$ sudo cp -R /media/mloveless/bkup/apt/sources.list/ /etc/apt/sources.list.d/
```

```
$ sudo apt update
```

```
$ sudo apt upgrade -y
```

```
$ sudo apt autoremove -y
```

- The commands above will restore pointers to the Dell Ubuntu repository, perform updates, and remove unused packages introduced during this entire process (none occurred during testing, this is to be sure).

General Applications

- Enable the firewall with the following command:

```
$ sudo ufw enable
```

- The prompt will respond with "Firewall is active and enabled on system startup".
- Some common applications to install include Google Chrome, Slack, and Zoom. Not all may be available via the normal Ubuntu repository, but you can download them via their respective sites with up-to-date installation instructions:
 - Google Chrome
 - Slack
 - Zoom
 - Checkout our Tools page for more potential items.

Usage of Java

Some applications used on Linux may require Java. The last open-source version of Oracle Java that was released was in January of 2019. All new versions since then require a paid/licensed subscription. Therefore GitLab no longer supports Oracle Java, and requires all team-members to use an open-source alternative like OpenJDK. Oracle periodically audits all downloads of Oracle Java and actively pursues companies that are out of compliance. The IT department therefore enforces a policy that will remove all instances of Oracle Java that are found on team-members machines

To ensure you are using the correct version, use the `java -version` command.

If OpenJDK is installed, the response will look similar to this:

```
shell
$ java -version
openjdk version "11.0.14.1" 2022-02-08
OpenJDK Runtime Environment (build 11.0.14.1+1-Ubuntu-0ubuntu1.20.04)
OpenJDK 64-Bit Server VM (build 11.0.14.1+1-Ubuntu-0ubuntu1.20.04, mixed mode, sharing)
```

If Oracle Java is installed, the response will look similar to this:

```
shell
$ java -version
java version "16.0.1" 2021-04-20
Java(TM) SE Runtime Environment (build 16.0.1+9-24)
Java Hotspot(TM) 64-Bit Server VM (build 16.0.1+9-24, mixed mode, sharing)
```

Most systems will be running either the OpenJDK version or Java will not be installed. If Java is not installed and you wish to install OpenJDK, follow the instructions for installation you received after running the `$ java -version` command. If you are running the Oracle Java version, follow the instructions here for Ubuntu, and for other distributions follow the instructions for package deletion and installation for your specific distribution.

Dell and Nvidia

Some GitLab team members have experienced issues with the Nvidia drivers on Dell, including battery drain due to sleep issues when the laptop is closed, random lockups when waking the laptop up, and so on. If this happens, consider the following steps:

- Examine the `/etc/default/grub` file, the line containing `GRUBCMDLINELINUXDEFAULT` may look like this:

```
GRUBCMDLINELINUXDEFAULT="quiet splash"
```

- Edit the line to look like this:

```
GRUBCMDLINELINUXDEFAULT="quiet splash memsleepdefault=deep"
```

- Then update grub:

\$ sudo update-grub

This has reported to work with latest versions of the Nvidia drivers (as of Dec 2019) so you can update the drivers.

Engineering/coding tools

- It's advised to have some sort of version manager for programming languages as what's supplied by repos is typically not sufficient
 - asdf is pretty good, and usable across many platforms
 - This project includes a large amount of plugins to control various packages as well as languages that you'd need
- Other common packages - this list includes things that always appear to be required no matter what realm of work you are in:
 - gcc
 - git
 - libssl-dev
 - make
 - zlib1g-dev
 - libreadline-dev Prior to Ubuntu 20.04
- Abide by our security practices:

Production engineering

- As a Production Engineer, we'll need some common tooling
 - AWS CLI
 - Azure CLI
 - Docker
 - Vagrant
 - VirtualBox
 - gcloud CLI
- Other important packages:
 - gnupg2
 - pwgen
 - sddaemon
 - yubikey-personalization
- If you chose to utilize asdf as mentioned above, install a few plugins to prep yourself to be ready to install a few tools in the future:
 - asdf plugin-add golang
 - asdf plugin-add kubectl
 - asdf plugin-add minikube
 - asdf plugin-add ruby
 - asdf plugin-add terraform
- Then you'll need to find out which version of the above needs to be installed
 - Check the appropriate repos or ask someone:
 - asdf install ruby 2.4.4
 - asdf install terraform 0.11.5

Development

- In development, we'll probably need the following packages installed:
 - cmake
 - g++
 - krb5
 - libkrb5-dev
 - libmysqlclient-dev
 - libpq-dev
 - libre2-dev
 - libsqlite3-dev
- If you chose to utilize asdf as mentioned above, install a few plugins to prep yourself to be ready to install a few tools in the future:
 - asdf plugin-add nodejs
 - asdf plugin-add postgres
 - asdf plugin-add ruby
- Then you'll need to find out which version of the above needs to be installed
 - Check the appropriate repos or ask someone:
 - asdf install ruby 2.4.4
 - asdf install nodejs 8.11.3

Enabling the fingerprint reader

Some of the authorized Dell laptops come with a fingerprint reader which is not supported by their official Ubuntu images. However, the fingerprint reader can be enabled using fprintd.

- Install the required software:

```
shell
sudo apt install fprintd libpam-fprintd
```

- Update the PAM configuration:

```
shell
sudo pam-auth-update
```

- Edit the /etc/pam.d/common-auth configuration to enable fingerprint authorization in the terminal and other password prompts. Find the auth [success=1 default=ignore] pamunix.so nulloksecure line and replace it with the following two lines.

```
shell
auth [success=2 default=ignore] pamfprintd.so max-tries=3 timeout=10
auth [success=1 default=ignore] pamunix.so nulloksecure
```

Note that max-tries is the number of fingerprint scans you can attempt until prompted for a password instead, and timeout is how long you have to scan your fingerprint before the authorization times out. You can configure these to your requirements.

- Enable fingerprint authorization on the login screen (sddm) by editing the /etc/pam.d/sddm

file and adding the following lines at the top of the file.

```
shell
auth    sufficient    pamunix.so tryfirstpass likeauth nullok
auth    sufficient    pamfprintd.so
```

- Finally, enroll your fingerprint with fprintd:

```
shell
fprintd-enroll $USER
```

Fingerprint login and authorization has now been enabled! Note that the sddm login screen has an unintuitive user interface. You'll be prompted with a password field as usual. Press enter and the screen will fade slightly. Now you can scan your fingerprint to log in.

Troubleshooting

Zoom screen sharing on GNOME on Wayland

In order to share the user's screen on GNOME on Wayland, Zoom uses a private screenshot API to chain successive screenshots into a stream. As GNOME 41, those private D-Bus APIs have been restricted to their intended callers for security reasons, so that hack no longer works..

As a workaround, you can use looking glass to set `global.context.unsafemode = true`. You can reinstate default security settings either by ending the session, or by running `global.context.unsafemode = true`.

Some caveats apply to this workaround.

- You are disabling a security setting. (This is no worse than a default Xorg session.)
- Zoom freezes for about 15 seconds when initiating a screen sharing session. Wait through the "application not responding" dialogue if it appears, and the "select display" prompt will appear.
- You can share whole desktops, but you cannot share individual application windows.
- Zoom freezes for about 5 seconds upon ending a screen sharing session.
- This workaround must be applied at each login.

Common issues

Here's a list of common situations that prove to be problematic on Linux.

- You'll want to ensure these components work as desired:
 - Audio through various types of headphones
 - Video capturing - Zoom video and Zoom screen sharing
 - Display - screen resolution or video card related issues
- When having problems with Okta under Linux, make sure to:
 - To use the latest Chrome (not Chromium) and your Yubi-Key or a phone without a custom ROM
 - Install SentinelOne and DirectStrike after your start as soon as possible

SECTION: tools-and-tips/mac.md

This page lists various tips and tools for macOS.

macOS tips

Here are some useful tips if you are new to macOS.

Even if you are not new, these keyboard shortcuts may come in handy.

If you are switching from Windows to macOS, these tips may be useful.

Taking screenshots and videos

On a macOS, there are a few screen shot key shortcuts that can be very helpful:

- command + shift + 3 - takes a screen shot of your entire screen(s - it will create an image for each screen you have)
- command + shift + 4 - allows you to select an area on the screen to take a screen shot (it will create one image of your selected area)
- command + shift + 5 - brings up a tool that allows you to do all of these things as well as create a video (with audio) of all or part of your screen.

In the case of command + shift + 3 and command + shift + 4, if you also hold your control key (so, command + shift + control + 3), your screen shot will be in your clipboard. Alternatively you can quickly select the screenshot preview in the bottom right corner and press command + c.

You can paste the screenshot from your clipboard into a lot of applications with command + v. GitLab supports pasting the screenshot into issue descriptions, comments and the Web IDE.

If you have a screen shot in your clipboard, you can open the Preview and then create an image from the clipboard using the File -> New from clipboard.

For screen recordings, you can use Quicktime Player to capture an area or the full screen. Open the application and, under the File menu, select New Screen Recording, or press command + control + n.

Screenshot productivity tips

To automatically compress your screenshots and move them to a designated directory, so your Desktop directory won't get cluttered with screenshot files, check this [GitLab Unfiltered blog post](#): One simple trick to make your screenshots 80% smaller.

Disabling macOS notification center

During a presentation or screen share, you might want to disable your notifications on macOS to prevent distractions or possible embarrassment.

The Notification Center can be quickly disabled by Option-Clicking the menu bar icon in the top

right of your screen.

This disables notifications until the next day.

Option-Click again to re-enable immediately.

Alternatively, click on the Notification Center icon, then scroll up to reveal the "Do Not Disturb" toggle.

If your laptop is a MacBook with a Touch Bar, note that you can assign a handy "Do Not Disturb" button on your Control Strip.

In System Preferences, navigate to Keyboard settings and click "Customize Control Strip..." to add this.

Locking your screen

On your macOS menu bar

1. Click on the Apple logo · menu
1. Click "Lock Screen"

Alternatively, you can use the keyboard shortcut ⌘+⌥+Q (ctrl+cmd+Q) or press Touch ID, if it's available on your Mac or Magic Keyboard.

On your macOS Touch Bar

1. Open System Preferences > Keyboard
1. Click Customize Control Strip
1. Drag the Screen Lock icon to the Touch Bar
1. Done!

Docker Desktop

Docker Desktop is a licensed tool and approved for usage and can be accessed via instructions in the Tech Stack.

To request access to Docker, please follow these steps:

1. Create an account on hub.docker.com using your @gitlab.com email. It is not possible to gain access with a non-company email.
1. Log an Access Request with the AccessChangeRequest template, detailing why alternatives are not an option (as noted above).
1. Once your manager approves the request, assign to the System Provisioner listed in the [Tech Stack] for this system.

Once your access has been provisioned, you will receive an email notification that you have been added to a specific dockerdesktop team. Following that, you can install Docker desktop, and login with the registered account.

The number of licenses is limited, so provisioning might take time, or might not be temporarily possible to gain access to the team.

If at some point in the future you do not want to use the product anymore, please file an

Access Change request, assign to provisioner listed in the Tech Stack so that we can assign the license to someone else.

The following open source alternatives appear to be maintained (as of 2025-02-28):

1. Rancher Desktop
1. Colima

Usage of Java

Some applications used on MacOS may require Java. The last open-source version of Oracle Java that was released was in January of 2019. All new versions since then require a paid/licensed subscription. Therefore GitLab no longer supports Oracle Java, and requires all team-members to use an open-source alternative like OpenJDK. Oracle periodically audits all downloads of Oracle Java and actively pursues companies that are out of compliance. The IT department therefore enforces a policy that will remove all instances of Oracle Java that are found on team-members machines.

To ensure you are using the correct version, use the `java -version` command.

If OpenJDK is installed, the response will look similar to this:

```
shell
$ java -version
openjdk version "18.0.1" 2022-04-19
OpenJDK Runtime Environment Zulu18.30+11-CA (build 18.0.1+10)
OpenJDK 64-Bit Server VM Zulu18.30+11-CA (build 18.0.1+10, mixed mode, sharing)
```

If Oracle Java is installed, the response will look similar to this:

```
shell
$ java -version
java version "16.0.1" 2021-04-20
Java(TM) SE Runtime Environment (build 16.0.1+9-24)
Java Hotspot(TM) 64-Bit Server VM (build 16.0.1+9-24, mixed mode, sharing)
```

Most systems will be running either the OpenJDK version or Java will not be installed. If Java is not installed and you wish to install OpenJDK, open the Self Service app located in your Applications folder and double-click on the app called Java 18 - Open Source.

If you are running the Oracle Java version, you will be automatically notified by the IT Department with a pop-up that looks like this

and it will be removed from your machine.

SECTION: tools-and-tips/mermaid.md

Mermaid diagrams

For the main Mermaid documentation please refer to the Tools and Tips page.

This page is intended to help out with some advanced layout options for Mermaid diagrams such as creating diagrams that are wider than the handbook main content area.

Gantt

<details>

<summary markdown="span">Code</summary>

md

gantt

title A Gantt Diagram

dateFormat YYYY-MM-DD

section Section

A task :a1, 2014-01-01, 30d

Another task :after a1 , 20d

section Another

Task in sec :2014-01-12 , 12d

another task : 24d

</details>

mermaid

gantt

title A Gantt Diagram

dateFormat YYYY-MM-DD

section Section

A task :a1, 2014-01-01, 30d

Another task :after a1 , 20d

section Another

Task in sec :2014-01-12 , 12d

another task : 24d

Flowchart (centered)

<details>

<summary markdown="span">Code</summary>

md

graph TD

A[Christmas] -->|Get money| B(Go shopping)

```
B --> C{Let me think}
C -->|One| D[Laptop]
C -->|Two| E[iPhone]
C -->|Three| F[fa:fa-car Car]
```

</details>

```
mermaid
graph TD
  A[Christmas] -->|Get money| B(Go shopping)
  B --> C{Let me think}
  C -->|One| D[Laptop]
  C -->|Two| E[iPhone]
  C -->|Three| F[fa:fa-car Car]
```

Sequence Diagram (right aligned)

<details>
<summary markdown="span">Code</summary>

```
md
sequenceDiagram
    Alice->>+John: Hello John, how are you?
    Alice->>+John: John, can you hear me?
    John-->>-Alice: Hi Alice, I can hear you!
    John-->>-Alice: I feel great!
```

</details>

```
mermaid
sequenceDiagram
    Alice->>+John: Hello John, how are you?
    Alice->>+John: John, can you hear me?
    John-->>-Alice: Hi Alice, I can hear you!
    John-->>-Alice: I feel great!
```

Gantt (wide scrollable)

<details>
<summary markdown="span">Code</summary>

```
md
gantt
    title A Gantt Diagram
    dateFormat YYYY-MM-DD
    section Section
```

```
A task      :a1, 2014-01-01, 30d
Another task :after a1 , 20d
section Another
Task in sec  :2014-01-12 , 12d
another task  : 24d
```

</details>

```
mermaid
graph TD
    title A Gantt Diagram
    dateFormat YYYY-MM-DD
    section Section
    A task      :a1, 2014-01-01, 30d
    Another task :after a1 , 20d
    section Another
    Task in sec  :2014-01-12 , 12d
    another task  : 24d
```

SECTION: tools-and-tips/onepassword-cli.md

Secrets used on laptops need to be stored and used securely. The reasons for this and basic concepts are discussed on the 1Password page. It's recommended to read that first if you haven't done so yet. This page covers user-friendly ways to integrate these basic principles into your workflow.

Standard command settings with aliases

Using an alias works well where a single, specific command should always use a single, specific set of secrets.

For example, you can define an alias in your shell profile to invoke glab with your PAT:

```
sh
alias glab="op run --env-file=$HOME/.gitlab-pat.env -- glab"
```

To verify the configuration, run `glab api version`. This should print the version of gitlab.com if the configuration succeeded.

```
sh
glab api version
{"version":"15.4.0-pre","revision":"3e84f577d51"}
```

Multi-environment usage with `.bashrc`

If you have a number of environments that you use frequently you can create functions to easily load secrets for the correct environment.

One example of this is below. It adds a 'loadenv' command to bash which automatically starts a subshell containing all the relevant environment variables, assuming the current directory is a git repository and an environment variable with a matching name is available in \$HOME/devenv/envfiles.

The modified \$PS1 checks for the presence of the 'ENVFILE' variable, and if it exists displays it in the shell prompt to show what environment file is loaded in this shell.

shell

Base PS1 + add (\$ENVFILE) in red if set

```
export PS1='\[e]0;\u@h: \w\a\${debianchroot:+($debianchroot)}\[\033[01;32m\]\u@h\[\033[00m\]:\[\033[01;34m\]\w\[\033[00m\]${[ -n "$ENVFILE" ] && echo -n " \e[91m($ENVFILE)\e[0m"}\$ '
```

alias project="git remote get-url origin | sed -e 's/^\.V//g' | sed -e 's/\.git\$/g'"

Environment file load

loadenv() {

ENVFILE=\$(project) op run --env-file="\$HOME/devenv/envfiles/\$(project).env" --no-masking -- /bin/bash

}

When added to .bashrc and sourced, it can then be used to run commands like this:

shell

username@my-laptop:~/code\$ cd example-project

username@my-laptop:~/code/example-project\$ loadenv

username@my-laptop:~/code/example-project (example-project)\$ curl -X POST --form token=\$PIPELINETOKEN --form ref=main https://gitlab.com/api/v4/projects/example-project/trigger/pipeline

{"id":1252595607,"iid":12960,...long output removed...

Then, once the secrets are no longer needed, you can exit the subshell and this will be visible in the prompt.

shell

username@my-laptop:~/code/example-project (example-project)\$ exit

exit

username@my-laptop:~/code/example-project\$

This allows maintaining a set of standard variables for a given git repository, regardless of the exact command(s) being run.

SECTION: tools-and-tips/other-apps.md

General security tip

Please review our acceptable use policy.

Some tools and extensions integrate into your various work accounts and will request certain permissions for them. Please always be cautious of the permissions requested by the application. For example a GitLab integration that only requests to be able to read your user profile can be appropriate, however an integration shouldn't be able to have read or write access the API. Similarly, you should not input a GitLab Personal Access Token linked to your work account in a third-party tool.

Here are some non-exhaustive guidelines:

- If the application requires credentials (password, access token, etc.) to any of your work accounts (GitLab, Google, Okta, etc.) please do not enter the credentials and remove the application
- If the application uses OAuth, make sure to only allow access to your account if the permissions required by the application are very minimal (such as accessing your public profile)

If the application requires anything such as read-only access to the API or anything that could grant access to confidential data please do not continue the authentication process and remove the application

- When installing something from an "app store" of some sort, look for the reviews, the number of downloads and especially for "badges" that show that the application was reviewed by the app store. For example the Chrome Web Store has a featured badge for popular trustworthy applications
- When in doubt, do not install the application

In any case, you are encouraged to use our Individual Use Software Request process prior to installing third party apps.

Internet browsers

Ad privacy

Sharing your screen to get your idea across can be very productive, but having personalized ads show up on a webpage may be undesirable.

Shut off interest based ads by setting your preferences.

Google Ad Settings, AdChoices

Browser extensions

In general, if a particular application or browser extension (sometimes called a plugin) is referenced in the handbook, it is considered "approved".

For example, 1Password is centered around the browser extension. Another application is Zoom, which has a scheduler extension. However, be sure to search for specific information about the application.

If you wish to use an extension not referenced in the handbook, consider the following before installing and using it:

- The extension should be work-related and help your overall productivity.
- The extension should be available from a reputable source, such as the browser's library of approved extensions.
- Ask. Feel free to ask your co-workers about good extensions, and if you have security or privacy concerns about an extension, ask the security team in security on Slack.

Some browser extensions are listed below

Adblockers

Adblockers are browser extensions that can block advertising, prevent user tracking, and include other security-related features.

A popular one recommended by the Security Team is uBlock Origin which can be installed for Chrome (uBlock Origin Lite).

One Tab

One Tab (Free) turns tabs into a list which can be sorted and exported.

SessionBox

SessionBox is a browser extension that helps you deal with multiple sessions.

It binds a particular session to a tab.

This is particularly useful for testing with different users in the same browser.

RSS Feed Reader

If you would like to receive daily notifications on newly opened issues, the Chrome extension RSS Feed Reader is an excellent tool for accomplishing this task.

After installing the extension, access the project page you are interested in following, under the project issues click on the "Subscribe to RSS feed" button which you can find in the top right corner of the page.

Flash (do NOT use)

Flash: Due to security flaws, we strongly recommend not using Adobe Flash.

Certainly do not install it on your local machine.

But even the Google Chrome plugin that lets you see embedded Flash content in websites can pose a security hazard.

If you have not already, go to your Chrome Flash Settings and disable Flash.

For further context, note that Google Chrome is removing Flash support soon, and while the plugin is better than a local install of Flash,

it still leaves vulnerabilities for zero-day attacks.

Prototyping in the browser

Sometimes you only need to capture small textual or visual changes in a web page as part of a bug report or a feature proposal.

You can use development tools that are usually built-in in most browsers which allow you to select and edit page element attributes as well as move around page elements like buttons or links.

You can also make the entire web page editable, using the `designMode` attribute, by typing `document.designMode="on";` in the development tools console or creating a bookmark by dragging the button below to your Bookmarks Bar.

```
<a href="javascript:document.body.contentEditable='true'; document.designMode='on'; void 0" class="btn btn-primary">Edit page</a>
```

Notes/writing

Keep in mind that some of the notes and writing tools make use of cloud servers on the back end, so do not put sensitive information into them.

Bear

Bear (Free) is a clean writing tool for notes and long-form writing. Ulysses \$5/month is also a great choice.

LanguageTool

As an alternative to Grammarly, we can combine LanguageTool in the browser with a local LanguageTool server .

This ensures that spell and grammar checking remains both convenient and confidential:

1. Install the LanguageTool browser extension.
1. Either install LanguageTool as a Homebrew service, or install a Docker engine and follow the setup instructions of any LanguageTool Docker image provider.
1. Configure the browser extension under Experimental settings > Local server.

Simplenote

Simplenote is a free, open source note taking app which is cross platform, syncs across all devices, and supports markdown.

Productivity

Similar to the extensions referenced